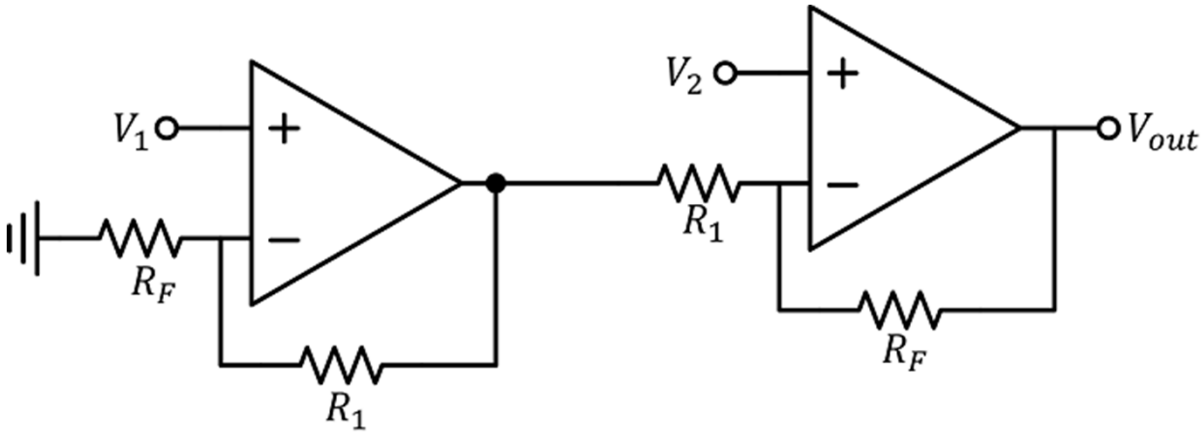
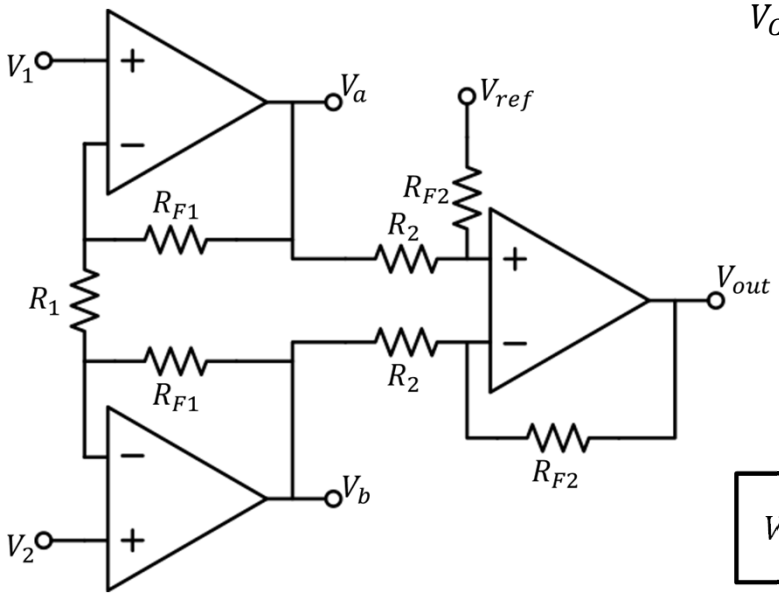
Amplificateur différentiel avec très grand impédance d'entrée
Differenzverstärker mit sehr hoher Eingangsimpedanz



$$V_{out} = \frac{R_1 + R_F}{R_1}(V_2 - V_1)$$

Ampli Différentiel avec très grand impédance d'entrée et une grande réjection de Mode Commun
Differenzverstärker mit sehr hoher Eingangsimpedanz und grosser Gleichtaktunterdrückung

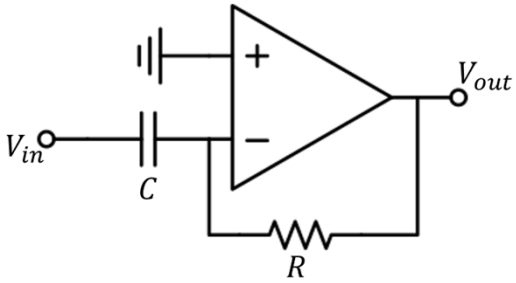


$$V_{CM} = \frac{V_1 + V_2}{2} \qquad V_d = V_1 - V_2$$

$$V_a = V_{CM} + \frac{R_1 + 2R_{F1}}{2R_1} V_d$$

$$V_b = V_{CM} - \frac{R_1 + 2R_{F1}}{2R_1} V_d$$

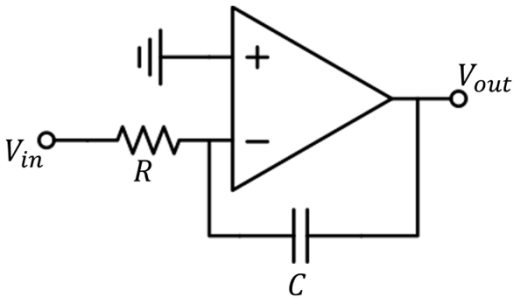
$$V_{out} = \frac{R_{F2}}{R_2}\frac{R_1 + 2R_{F1}}{R_1}(V_1 - V_2) + V_{ref}$$



Différentiateur / *Differentiator*

$$V_{out} = -RC\frac{dV_{in}}{dt}$$

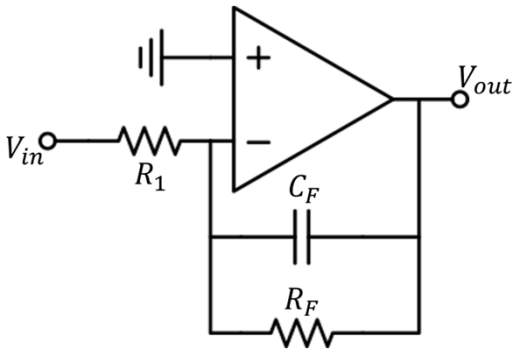
$$V_{out} = -RCj\omega V_{in}$$



Intégrateur / Integrator

$$V_{out} = -\frac{1}{RC}\int_0^t V_{in}\,d\tau$$

$$V_{out} = -\frac{1}{RCj\omega}V_{in}$$

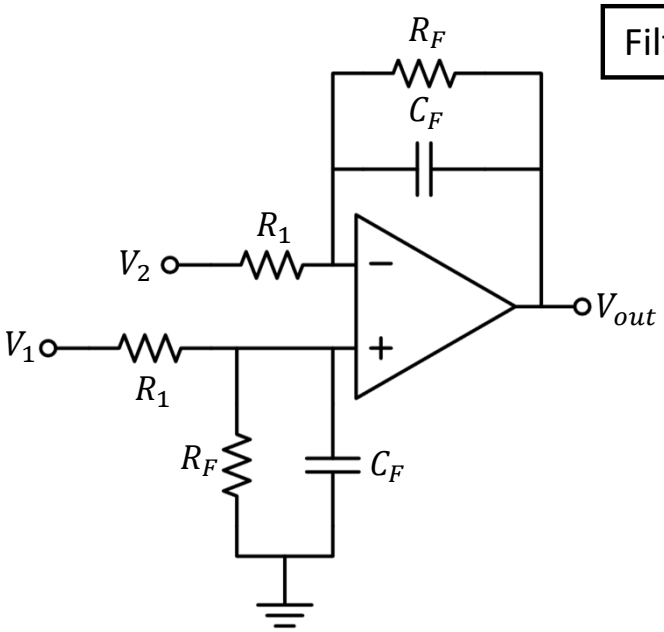


Filtre Passe-bas / Tiefpassfilter

$$V_{out} = -V_{in}\frac{R_F}{R_1}\frac{1}{1+j\omega CR_F}$$

Intégrateur / Integrator

$$V_{out} = -\frac{1}{R_1 C}\int_0^t V_{in}\,d\tau, \omega \gg \frac{1}{CR_F}$$



Filtre Passe-bas différentiel / Differenztiefpassfilter

$$V_{out} = \frac{R_F}{R_1}\frac{1}{1+j\omega CR_F}(V_1 - V_2)$$